

1-Task Title: Function Check (One-to-One and Onto)

Code:

```
a=int(input("Total member of Set A-> "))
b=int(input("Total member of Set B-> "))
print("Taking Set A")
A=list(map(int,input().split()))
A=set(A)
print("Taking Set B")
B=list(map(int,input().split()))
B=set(B)
C=[]
on=True
for i in A:
    F=i**3+i**2-i-1
    if F not in B:
        print(f"It is not a Function As there is no element in co-domain for domain {i}\n")
        on=False
        break
    C.append(F)
if on:
    C=set(C)
    print("One-to-One-> Yes") if len(A)==len(C) else print("One-to-One-> No")
    print("Onto-> Yes") if B==C else print("Onto-> No")
```

Output:

```
Total member of Set A-> 3
Total member of Set B-> 3
Taking Set A
1 2 3
Taking Set B
0 9 32
One-to-One-> Yes
Onto-> Yes
PS F:\RUET CSE LAB\Discrete Mathematics 2102\5. 03-02-2024>
```

Discussion:

In this program I have tried to implement using domain and co-domain if it is one to one or onto function. If the size of the domain is equal to the size of range, then it is a One-to-One Function. Again, if all the co-domain has at most one domain, then it is called an Onto Function. Furthermore, if a domain is unable to find a co-domain then it is not a function.

2-Task Title: Summation

Code:

```
L1=int(input("Lower bound of first summation-> "))
U1=int(input("Upper bound of first summation-> "))
L2=int(input("Lower bound of second summation-> "))
U2=int(input("Lower bound of second summation-> "))
sum=0
for i in range(L1, U1+1):
    for j in range(L2, min(i, U2)+1):
        sum+=(5*i*2*j)
print(sum)
```

Output:

```
Lower bound of first summation-> 2
Upper bound of first summation-> 5
Lower bound of second summation-> 1
Lower bound of second summation-> 4
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PS F:\RUET CSE LAB\Discrete Mathematics 2102\5. 03-02-2024>
```

Discussion:

In this code segment I have implemented a summation function like $\sum \sum (5i \times 2j)$. First Summation is supposed outer part, and second summation is supposed to inner part. Then created a nested 'for loop' for the outer part and inner part between range and calculated the result from it.